

1 Quantum shoving

We want to explore how string shoving works when given a quantum mechanical treatment.

The picture is that of a single string, with wave function $\psi(t, x)$ being pushed by a potential V , which can come from another single string, or a multiplet. This is different than the classical treatment, which is only string-on-string interactions.

2 Two-site toy model

We consider a toy setup. A two-site lattice, labelled a_1 and a_2 :

$$|a_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |a_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1)$$

Our string starts in a_1 , and therefore

$$|\psi(0)\rangle = |a_1\rangle. \quad (2)$$

In sec. 2.1 we briefly sketch the behaviour when no potential is added, mostly to set up notation and to be able to compare with the case of having one, and in sec. 2.2 we explore what happens when a potential is added. This turns out to be equivalent to the two-state Rabi model. This is valuable when we do the time evolution in sec. 2.3.

2.1 No potential

In the absence of a potential, the string would be able to jump from one lattice site to the other governed by an amplitude, J , connected to the kinetic energy of the string. The Hamiltonian is:

$$H = \begin{pmatrix} 0 & -J \\ -J & 0 \end{pmatrix}. \quad (3)$$

With eq. (2) we can find the probability to be in $|a_2\rangle$ after some time t . The energy eigenvalues are trivially $E = \pm J$. The corresponding normalized eigenstates of the Hamiltonian are the symmetric and antisymmetric superpositions of the two lattice sites:

$$|\phi_1\rangle = \frac{1}{\sqrt{2}} (|a_1\rangle + |a_2\rangle), \quad E = -J, \quad (4)$$

$$|\phi_2\rangle = \frac{1}{\sqrt{2}} (|a_1\rangle - |a_2\rangle), \quad E = +J. \quad (5)$$

We express the initial state as a superposition of these eigenstates:

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|\phi_1\rangle + |\phi_2\rangle). \quad (6)$$

The time evolution of the state is then given by:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} (e^{iJt}|\phi_1\rangle + e^{-iJt}|\phi_2\rangle). \quad (7)$$

Rewriting this back in the position basis gives:

$$|\psi(t)\rangle = \cos(Jt)|a_1\rangle - i\sin(Jt)|a_2\rangle. \quad (8)$$

Thus, the probability of finding the string at site $|a_2\rangle$ at time t is:

$$P_{a_2}(t) = |\langle a_2|\psi(t)\rangle|^2 = \sin^2(Jt). \quad (9)$$

This describes coherent oscillations of the string between the two sites with a frequency determined by J , in accordance with what one would expect.

We can also examine the expectation value of the total energy, which in this case is purely kinetic since no potential is present. In the initial state we have:

$$\langle\psi(0)|H|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \left[\begin{pmatrix} 0 & -J \\ -J & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] = 0. \quad (10)$$

The expectation value of the kinetic energy is zero in the initial state because the state is an equal superposition of the two energy eigenstates. This does not mean that we violate Heisenberg uncertainty. The variance in energy is:

$$(\Delta E)^2 = \langle\psi(0)|H^2|\psi(0)\rangle - (\langle\psi(0)|H|\psi(0)\rangle)^2 = \langle a_1|J^2 \cdot \mathbb{I}|a_1\rangle - 0^2 = J^2 \Leftrightarrow \Delta E = J, \quad (11)$$

reflecting that the initial state is a superposition between the two energy eigenstates from above. So while the position is fully localized in space, it has maximum uncertainty in momentum as we should require.

2.2 Adding a potential to the toy model

We now add a potential, V , to site a_1 . The Hamiltonian becomes:

$$H = \begin{pmatrix} V & -J \\ -J & 0 \end{pmatrix}. \quad (12)$$

Again we start with eq. (2) and want to find the probability to be in $|a_2\rangle$ after some time t . Again we can find the eigenvalues (E):

$$\det(H - E\mathbb{I}) = \det \begin{pmatrix} V - E & -J \\ -J & -E \end{pmatrix} = 0, \quad (13)$$

which gives

$$E^2 - VE - J^2 = 0 \Rightarrow E_{\pm} = \frac{V}{2} \pm \sqrt{\left(\frac{V}{2}\right)^2 + J^2}. \quad (14)$$

The energy levels are no longer symmetric around zero due to our introduction of V . As it turns out, this symmetry breaking introduces an inertia which makes it harder to move away from a_1 , just like the spontaneous symmetry breaking of the Higgs mechanism introduces mass. In our case, however, the symmetry breaking is not spontaneous, but explicit. We will return to this point.

Next, we find the eigenstates. For each eigenvalue, E_i , the eigenstate, $|\phi_i\rangle$ satisfies:

$$(H - E_i \cdot \mathbb{I}) |\phi_i\rangle = 0, \quad |\phi_i\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}_i \quad (15)$$

Expanding, we get:

$$\begin{pmatrix} V - E_i & -J \\ -J & -E_i \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}_i = 0. \quad (16)$$

Solving for c_1 and c_2 and normalizing, gives the normalized eigenstates:

$$|\phi_i\rangle = \frac{1}{\sqrt{1 + \lambda_i^2}} \begin{pmatrix} 1 \\ \lambda_i \end{pmatrix} \text{ with } \lambda_i = \frac{V - E_i}{J}. \quad (17)$$

Before going to time evolution, it is instructive to look at the eigenstates when the potential is large $V \gg J$, to reinforce the inertia analogy from before. We expand the square root in the energy eigenvalues ($\sqrt{1 + \epsilon} \approx 1 + \epsilon/2 + \dots$):

$$E_{\pm} = \frac{V}{2} \pm \frac{V}{2} \sqrt{1 + \frac{4J^2}{V^2}} \approx \frac{V}{2} \pm \frac{V}{2} \left(1 + \frac{2J^2}{V^2}\right), \quad (18)$$

so we get:

$$E_+ \approx V + \frac{2J^2}{V}, \quad E_- \approx -\frac{2J^2}{V}. \quad (19)$$

This makes λ_i in the eigenvectors:

$$\lambda_+ \approx \frac{2J}{V} \rightarrow 0, \quad \lambda_- \approx \frac{V}{J} + \frac{2J}{V} \rightarrow \infty. \quad (20)$$

With $V \gg J$, the normalized eigenvectors are therefore:

$$|\phi_+\rangle \approx \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |a_1\rangle, \quad |\phi_-\rangle \approx \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |a_2\rangle. \quad (21)$$

Recall, that with no potential added, the normalized eigenstates of the Hamiltonian were the symmetric and anti-symmetric superpositions of the two lattice sites, eq. (4) and eq. (5). With a large potential, they are now localized to the two lattice sites. This is what one would classically expect if the effect of a large potential is to induce inertia.

2.3 Time evolution: equivalence to a detuned two-level system

While it is possible to proceed with a direct derivation of $P_{a_2}(t)$ using the Hamiltonian in eq. (12), the algebra quickly becomes cumbersome. Instead, we recognize that our system is mathematically equivalent to a well-known two-level system, described by the two-level Rabi model Hamiltonian:

$$H = \begin{pmatrix} \Delta & -J \\ -J & -\Delta \end{pmatrix}, \quad (22)$$

where we have introduced $\Delta = V/2$ by shifting the zero-point of energy. This transformation does not affect the physical dynamics.

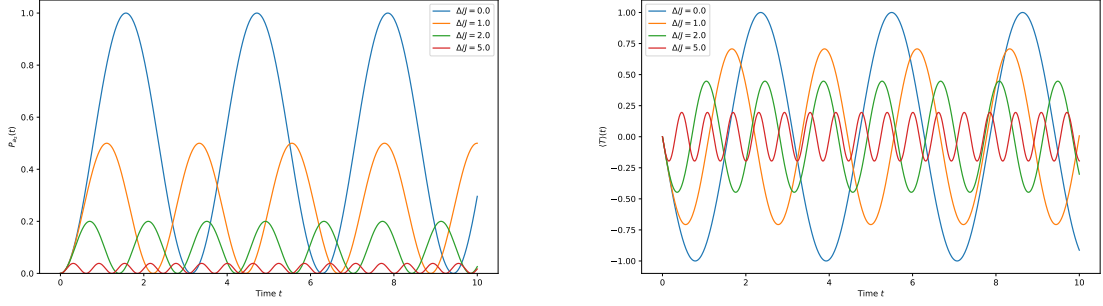


Figure 1: Transition probability and kinetic energy in the two-state system as function of time, with varied potential Δ/J .

The eigenvalues are now trivially:

$$E_{\pm} = \pm\sqrt{J^2 + \Delta^2}. \quad (23)$$

Derivation of the eigenstates is done in e.g. Sakurai.

$$|\phi_+\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad |\phi_-\rangle = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}, \quad \text{with } \tan(2\theta) = J/\Delta. \quad (24)$$

We can write $|a_1\rangle$ and $|a_2\rangle$ in energy eigenstates as:

$$|a_1\rangle = \cos(\theta)|\phi_+\rangle - \sin(\theta)|\phi_-\rangle, \quad |a_2\rangle = \sin(\theta)|\phi_+\rangle + \cos(\theta)|\phi_-\rangle, \quad (25)$$

The time-evolved state is then:

$$|\psi(t)\rangle = \cos \theta e^{-iE_+t}|\phi_+\rangle - \sin \theta e^{-iE_-t}|\phi_-\rangle. \quad (26)$$

We can now calculate $P_{a_2}(t)$:

$$\langle a_2|\psi(t)\rangle = \cos \theta e^{-iE_+t} \sin \theta + (-\sin \theta) e^{-iE_-t} \cos \theta = \sin \theta \cos \theta (e^{-iE_+t} - e^{-iE_-t}). \quad (27)$$

Thus, the probability is:

$$P_{a_2}(t) = |\langle a_2|\psi(t)\rangle|^2 = \sin^2 \theta \cos^2 \theta \cdot |e^{-iE_+t} - e^{-iE_-t}|^2. \quad (28)$$

Using:

$$|e^{-iE_+t} - e^{-iE_-t}|^2 = 4 \sin^2 \left(\frac{(E_+ - E_-)t}{2} \right), \quad \text{and } E_+ - E_- = 2\sqrt{J^2 + \Delta^2} \quad (29)$$

We obtain:

$$P_{a_2}(t) = \sin^2(2\theta) \cdot \sin^2 \left(\sqrt{J^2 + \Delta^2} \cdot t \right). \quad (30)$$

Finally, from the definition of θ , we have:

$$\sin(2\theta) = \frac{\tan(2\theta)}{\sqrt{1 + \tan^2(2\theta)}} = \frac{J}{\sqrt{J^2 + \Delta^2}}, \quad (31)$$

so finally:

$$P_{a_2}(t) = \frac{J^2}{J^2 + \Delta^2} \cdot \sin^2 \left(\sqrt{J^2 + \Delta^2} \cdot t \right). \quad (32)$$

The probability for different Δ/J is shown in fig. 1 (left). It is seen clearly that when Δ , and thus V , is large, wrt. J the probability to transition vanishes, while the frequency increases. In fig. 1 (right) the expectation value of the kinetic energy $\langle T \rangle$ is shown, calculated analogously.

The kinetic term in this system represents not classical kinetic energy, but rather the ability of the string to transition between lattice sites. The time-dependent expectation value is given by:

$$\langle T \rangle(t) = -\frac{J^2}{\sqrt{J^2 + \Delta^2}} \cdot \sin \left(2\sqrt{J^2 + \Delta^2} \cdot t \right). \quad (33)$$

This expression reveals two key features: as Δ increases, the *amplitude* of kinetic activity decreases, while the *frequency* increases. Physically, a large potential suppresses transitions, effectively “freezing” the system in its initial state. This dynamical suppression can be interpreted as the emergence of an *effective mass*, where the system resists motion in analogy to classical inertia.

A dimensionally consistent definition of the effective mass is naturally provided by the characteristic energy scale of the system:

$$m_{\text{eff}} = \sqrt{J^2 + \Delta^2}. \quad (34)$$

In the limit $\Delta \gg J$, this reduces to $m_{\text{eff}} \approx \Delta$, indicating that the potential directly governs the dynamical inertia. The suppression of both the transition probability $P_{a_2}(t)$ and the kinetic expectation $\langle T \rangle(t)$ can thus be understood as a consequence of the ratio J/m_{eff} , reflecting how the coupling strength (J) competes with the inertial mass scale.

The purpose of this toy model is not realism, but control: it allows us to identify which features of string shoving arise from quantum dynamics alone, before adding spatial structure or numerics.

In the remainder of the project, this two-state system will be generalized to multi-site lattices and continuous wavepackets, where analytical solutions are no longer available and numerical time evolution becomes essential.”

3 Project plan and milestones

The project is organized into successive phases. The initial phase is tightly specified and focuses on establishing analytical and numerical control. Later phases are progressively less prescriptive and allow for exploration depending on progress.

3.1 Phase I: Two-site quantum toy model

Goal: Establish full analytical and numerical control of the two-site model introduced in the brief. This model serves as the reference point for all later generalizations.

Week 1: Analytical understanding of the two-site system

Tasks:

- Reproduce all derivations in Sections 2–4 of the project brief independently.
- Verify limiting cases explicitly:
 - $V = 0$
 - $V \gg J$
- Explore alternative initial conditions, such as superpositions of $|a_1\rangle$ and $|a_2\rangle$, and compute the resulting time evolution.

Deliverables:

- A short written note (2–3 pages) summarizing the two-site model, its spectrum, and its time evolution.
- Explicit expressions for transition probabilities and expectation values used later as benchmarks.

Week 2: Numerical implementation of the two-site model

Tasks:

- Implement time evolution for the two-site Hamiltonian using a numerical method (matrix exponential or Crank–Nicolson).
- Compute:
 - Transition probability $P_{a_2}(t)$
 - Expectation value of the kinetic operator $\langle T \rangle(t)$
- Compare numerical results to analytical expressions for multiple values of V/J .

Checks:

- Conservation of norm
- Convergence with respect to time step
- Agreement with analytical results to controlled numerical precision

Deliverables:

- Plots reproducing analytical curves.
- Short technical note documenting numerical choices and validation tests.

Week 3: Parameter exploration and robustness

Tasks:

- Perform systematic scans over V/J and J .
- Quantify how transition amplitudes and kinetic activity depend on these parameters.
- Identify regimes where transitions are strongly suppressed.

Deliverables:

- Parameter-scan plots.
- Tabulated reference values to be used for comparison with lattice and continuum models.

3.2 Phase II: One-dimensional lattice dynamics

Goal: Extend the two-site model to a spatially resolved one-dimensional lattice and study time-dependent Schrödinger evolution where analytical solutions are no longer available.

Tasks:

- Implement a one-dimensional lattice Schrödinger solver with nearest-neighbor hopping.
- Validate the solver using known test cases:
 - Free wavepacket propagation
 - Scattering off a static potential barrier
- Model a localized external potential representing interaction with another string.
- Compute reflection, transmission, and momentum-transfer observables.

Deliverables:

- Verified numerical solver with documented convergence tests.
- Time-dependent wavefunction and momentum-space distributions.

3.3 Phase III: Two-string scattering and cross sections

Goal: Formulate string-string interaction as an effective scattering problem and extract cross sections.

Tasks:

- Define a scattering setup in one dimension, and optionally extend to two dimensions.
- Identify relevant observables:
 - Transmission and reflection coefficients

– Effective scattering cross sections

- Derive analytical estimates using WKB methods where applicable.
- Compute scattering observables numerically and compare to analytical approximations.

Deliverables:

- Parameter-dependent scattering probabilities.
- Quantitative comparison between WKB and numerical results.

3.4 Phase IV: Extensions and further developments

Goal: Explore extensions of the baseline model depending on progress and results obtained in earlier phases.

Possible directions include:

- Extension to two transverse dimensions.
- Time-dependent interaction potentials.
- Ensemble averaging and robustness of observables.
- Reduction of quantum dynamics to effective interaction rules.

The precise scope of this phase will be determined jointly during the project.